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This edit has 3,727 reading words and failed the required 4,500-word minimum. It did not reach the post-polish semantic audit or targeted repair. It is not an accepted publication. The delivered fallback retains the original draft.

Epistemic Modals and Information-State-Sensitive Evaluation

Abstract

This paper defends an adequacy condition on the semantics of epistemic modals: such a semantics must make modal evaluation sensitive to information states, not only to worlds. The case rests on two commitments of dynamic semantics and one discourse contrast. The commitments are that conjunction receives a sequential interpretation and that might is interpreted as a consistency test. The contrast is that once the information that it is not raining has been added, it is no longer consistent with the information that it might be raining, while the opposite order raises no problem. The argument treats this contrast as a sign. It is explained when conjunction updates information in sequence and might tests consistency against the resulting state, and that explanation runs through information states. Two critical questions are examined: whether the contrast is reliable evidence, and whether the commitments support the condition rather than evaluation by worlds alone. Both receive affirmative answers, with limits. The argument is conditional on its premises. It does not deny worlds a role. And a single contrast does not show that every epistemic modal behaves in the same way. The result is a requirement of deliberately limited reach.

1. Introduction

Epistemic modals such as might are standardly assessed by asking what a semantics should evaluate them against. This paper argues for one answer in the form of an adequacy condition: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The claim concerns what any account must accommodate. It is not a proposal for a particular formal system.

Two features of this condition shape everything that follows. First, it does not say that worlds have no role in modal evaluation. It says that worlds alone are not enough and that information states must also figure in it. Second, it is a demand on an account. It is not a report that some account meeting the demand has been independently established, and it is not a generalization about how every epistemic modal behaves.

The condition is supported by two premises drawn from dynamic semantics. The first is a pair of semantic commitments: dynamic semantics gives conjunction a sequential interpretation and interprets the might-operator as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The second is a contrast in discourse. Once the information that it is not raining has been added, it is no longer consistent with the information that it might be raining; in the opposite order there is no problem, so order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The inference from these premises to the condition is an argument from a sign. The order contrast is evidence of how modal evaluation works because the commitments explain it, and the explanation they give runs through information states.

The paper proceeds as follows. Section 2 sets out the two semantic commitments and what each does and does not say. Section 3 states the order contrast and fixes its scope. Section 4 reconstructs the inference from the commitments and the contrast to the adequacy condition. Section 5 develops objections from the two critical questions that bear on that inference, and from questions about its scope, and replies to them. Section 6 states the limits of the support. Section 7 places the argument in the recent literature and fixes the scope of any claim to novelty. Section 8 concludes. The aim is to make one inference explicit and to show exactly how far it reaches.

2. Two Commitments of Dynamic Semantics

The first premise consists of two semantic commitments of dynamic semantics: conjunction receives a sequential interpretation, and the might-operator is interpreted as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The commitments concern different expressions, so they are best stated separately. Neither feature should be assigned to the other expression. Sequence is a property of how conjunction is interpreted, and consistency testing is a property of how might is interpreted. Neither needs to be expanded into a further semantic rule for the argument to go through.

The first commitment concerns conjunction, which is interpreted sequentially. As stated, the commitment says nothing about additional stages, outcomes or rules for how a particular conjunct changes an information state. It also says nothing on its own about what follows from any particular ordering. Its contribution is limited: it makes the order in which conjoined material is processed part of conjunction's interpretation (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). That is enough for the argument, which needs only that sequence is part of what conjunction contributes.

The second commitment concerns might, which is a consistency test. This characterization does not say how the test operates in further detail, and it gives might no other function. Even so, it ties modal evaluation explicitly to information. On this commitment, what matters for a might-claim is whether it remains consistent with the information state against which it is tested (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). A consistency test needs something to test against, and on this treatment that something is an information state.

Taken together, the two commitments shape how the account handles epistemic-modal claims. Sequence belongs to the interpretation of conjunction, and consistency with information belongs to the interpretation of might. The argument depends on this pairing. If information arrives in sequence

and might is tested against the information available at its point in the sequence, then the order in which information arrives can affect the outcome of the test. The scope of the pairing is equally definite. As presented, the commitments make no broader claim about all epistemic modals, about other modal operators, or about other properties of conjunction (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

3. The Not-Raining Order Effect

The second premise is a contrast between two orders in which information is added in discourse. In the first order, the information that it is not raining comes before the information that it might be raining. Once the not-raining information has been added, it is no longer consistent with the information that it might be raining. In the opposite order there is no problem. The source concludes from this contrast that order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

The contrast belongs to the sequence, not to the two pieces of information taken in either order. The consistency result attaches to the order in which the not-raining information comes first, and nothing in the description says that the same result arises whichever item comes first. The two orders also have different outcomes. One produces a consistency problem and the other does not, and the contrast needs both clauses. Without the second clause there is no difference between orders. Without the first there is nothing for the opposite order to differ from (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

The datum is narrow, and each of its parts has a limited scope. The first clause concerns only whether the might-information remains consistent after the not-raining information has been added. The second says only that the reverse order presents no problem; it gives no further outcome for that sequence. The phrase "order matters in discourse" names this particular contrast. It adds no general rule and makes no claim about other sequences, other kinds of information, other epistemic modals or other examples. As stated, the datum is that whether a problem arises in this pair of orders depends on which of the two pieces of information comes first (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

4. Argument: From the Contrast to the Adequacy Condition

The inference has the form of an argument from a sign. It has two premises: the commitments of Section 2 and the contrast of Section 3 . The contrast functions as the sign, and the commitments explain why it is a sign of something about modal evaluation rather than an isolated fact about two sentences. The reconstruction below proceeds in three steps. The first connects the commitments to the information state. The second connects the information state to the contrast. The third draws the conclusion that the contrast indicates.

The first step links the two commitments. If might is a consistency test, then the outcome for a might-claim depends on the information state against which it is tested. If conjunction is sequential, then the information state at a given point depends on what has been added before that point and in what order

(Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Together, these give a definite consequence. The outcome of the consistency test for a might-claim can differ depending on what information has already been added when the claim is reached.

The second step applies this consequence to the datum. When the not-raining information comes first, the state against which the might-information is tested already contains that information, and the might-information is no longer consistent with it. When the order is reversed, the might-information is tested before the not-raining information is added, and nothing goes wrong (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The commitments therefore predict a difference between the two sequences, and the datum records one. On these commitments, the order effect is what one should expect.

The third step draws the conclusion. The order effect is explained once evaluation runs against a sequentially updated information state, and the explanation works only because the consistency test is applied to such a state. Because the contrast is explained in this way, it counts as evidence about how modal claims are evaluated. Specifically, it is evidence that information states, and not worlds alone, enter into that evaluation. This is the requirement stated in Section 1 (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The step does not claim that worlds have been shown to play no part. What the explanation needs is that information states play a part, and that is all the conclusion asserts.

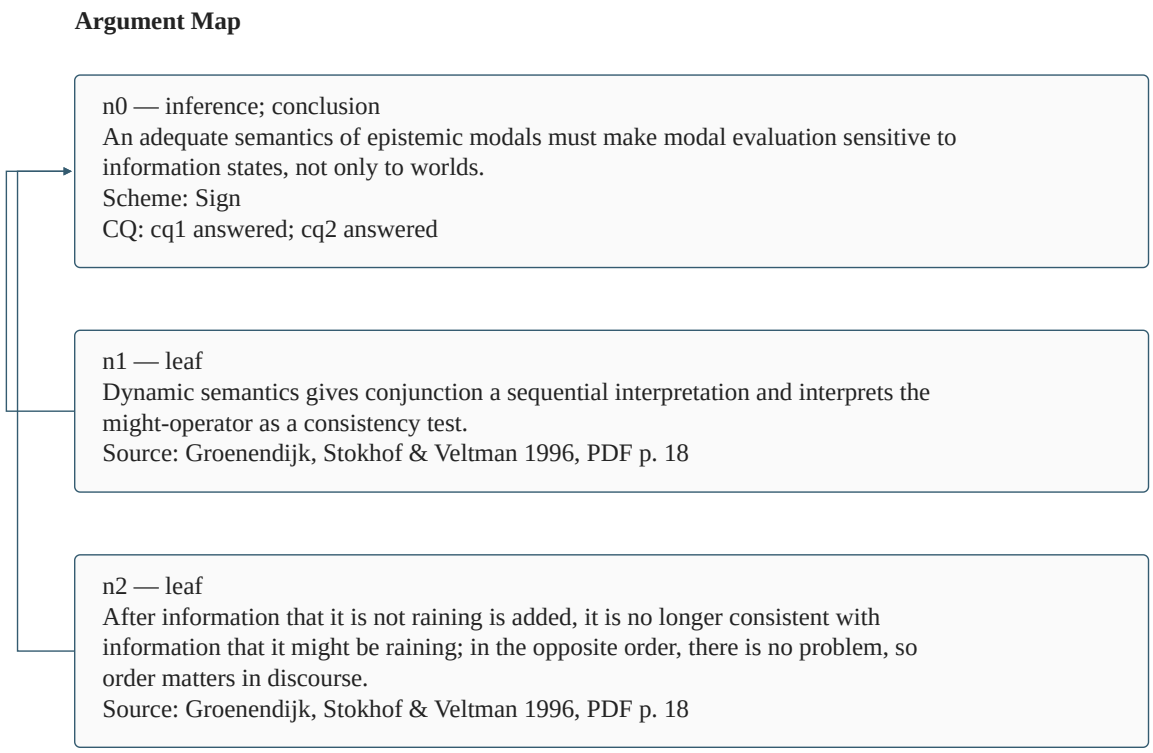


Figure 1. Route premises, inference schemes and conclusion. Arrows follow premise-to-conclusion links; CQ labels report recorded answers.

Support in this argument runs in one direction only, from the commitments and the order effect to the requirement. Nothing in the reconstruction suggests that the requirement independently proves the commitments, or that accepting the requirement would by itself fix how conjunction or might should be interpreted. The commitments and the contrast are the grounds. The requirement is what they support.

5. Objections and Replies

Two critical questions bear directly on the inference. The first asks whether the order contrast is reliable as evidence. The second asks whether the semantic commitments really support the adequacy condition. Further objections concern how far a single contrast can reach, whether the condition sidelines worlds, and whether the argument assumes what it sets out to show. Each is taken in turn.

The first objection targets the sign. A contrast between two discourse orders, it says, is a fact about discourse. It might reflect how speakers sequence information rather than how might is evaluated, and if so it would be unreliable as evidence about modal semantics. On this view, Section 4 moves too quickly from a discourse datum to a claim about modal evaluation.

The reply is that the contrast is reliable as evidence that epistemic-modal evaluation is sensitive to information states, and the reason is the way it is explained. The order effect is explained when conjunction updates information sequentially and might tests consistency (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). On that explanation, the discourse datum and the modal evaluation are not separate matters. The datum arises because might is tested for consistency against information that has accumulated in sequence. The objection is right that a discourse contrast taken alone would not reveal how a modal is evaluated. The contrast is not taken alone here, however. It is taken together with the commitments that explain it, and it is reliable as evidence only to the extent that it is so explained.

The second objection grants the explanation but questions what it supports. The commitments describe one semantic treatment of conjunction and might. The objection asks why they should support the adequacy condition rather than leaving open evaluation by worlds alone.

The reply is that the commitments do support the condition, and they do so through the role they give the information state. The might-operator is treated as a consistency test on an information state, and adding information in a different sequence changes whether the might-claim remains consistent (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The explanation of the order effect therefore depends on an information state that changes as information is added. An account that offered evaluation by worlds alone would not be using this explanation, because the explanation's key element is the state against which might is tested. This reply rests only on the stated commitments and the stated contrast. It does not rely on any further argument that worlds alone cannot do the work, and none is offered here.

A third objection concerns reach. One pair of sentences about rain, it says, cannot license a conclusion about the semantics of epistemic modals in general. The reply is to accept the premise and deny that the conclusion requires more. The contrast supports the condition, but it does not by itself establish that every epistemic modal behaves in this way (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The condition does not claim that every epistemic modal does. It is a demand that an adequate semantics be able to make information states matter to evaluation, and a single well-explained case is relevant to that demand. If the condition were recast as a universal generalization about epistemic modals, the objection would succeed. As formulated, it does not.

A fourth objection is that the condition dismisses worlds, which seem indispensable to modal semantics. The reply is that the condition does no such thing. Its wording is that evaluation must be sensitive to information states, not only to worlds, and it does not say that worlds have no role. Nothing in the reconstruction in Section 4 requires removing worlds from evaluation. What it requires is that information states also figure in it, since the consistency test is applied against such a state.

A fifth objection alleges circularity. The commitments, it says, already build information states into the semantics of might, so using them to support a requirement about information states assumes the conclusion. The reply distinguishes the commitments from the requirement. The commitments describe how one account interprets conjunction and might. The requirement states a condition on any adequate account. The order contrast connects the two: it is a datum that the commitments explain, and the commitments explain it by way of information states. So the argument does not simply assume the requirement. It holds that a datum is explained in a way that makes information states matter. The direction of support is from commitments and datum to requirement, as Section 4 stated. The objection is right that the argument stands or falls with the commitments, and Section 6 states that dependence as a limit.

6. Limits

The first limit is that the inference is conditional. It holds if conjunction is interpreted sequentially, if might tests consistency, and if the described order contrast obtains (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). None of these premises is independently established here. Both are taken from the cited dynamic account and are endorsed there. The argument adds an explicit reconstruction of how they bear on the adequacy condition. It does not add independent grounds for the premises.

The second limit concerns what the reconstruction leaves out. The cited account gives its own derivation of the order contrast. That derivation is not reproduced as part of this route, which takes the contrast as a premise attributed to the source. The source-support record below shows which route node each citation supports and with what attribution and stance. That record documents recorded support. It is not independent verification of the cited claims.

Source Support

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n1 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

n2 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

Figure 2. Frozen source support by route node, page citation, attribution and stance. Recorded support is not independent verification.

The third limit is the reach of the datum. The contrast shows how two specified sequences differ, and on its own it does not show that every epistemic modal behaves this way. It supports no conclusion about other sequences, other kinds of information, other modal expressions or other modalities (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The condition is formulated to stay within this reach. It is a demand on an adequate semantics, not a universal generalization about epistemic modals.

Taken together, these limits fix the size of the contribution. The argument makes explicit how one pair of semantic commitments and one discourse contrast bear on what a semantics of epistemic modals must accommodate. It shows that, given these premises, the contrast is a reliable sign and the commitments support the requirement. It does not establish the premises, survey further data, or adjudicate among accounts that might meet the requirement in different ways. Its scope is exactly as wide as this support.

7. Related Work and Scope of the Novelty Claim

A bounded search was conducted for recent literature bearing on this argument. It was a ranked search of indexes and public sources covering 2018–2026, run on 28 October 2026. Its topics included contextualism, relativism, expressivism, dynamic semantics, epistemic contradictions, probability operators, retraction and assessment sensitivity, and its queries included "epistemic modals dynamic semantics", "epistemic contradiction static dynamic semantics order" and "epistemic modals information sensitivity static semantics" (Literature-currency search 2026, 2018–2026 search). The sweep selected 25 recent works and obtained 17 exact full texts, together with three previously

missing originals. Eight selected recent works and 24 originals were unavailable. The search record notes that older foundational chapters, complete books and recent empirical reviews remain material access gaps, and that the search was not an exhaustive review (Literature-currency search 2026, 2018–2026 search). Nothing below should be read as a claim of exhaustive absence or as a certification of novelty.

Measured against this search, the thesis is partially anticipated by the read works. Information-sensitive approaches to epistemic modals already exist. A recent survey chapter, read in an author manuscript dated January 2025 that excerpts chapters of a 2024 book and is not the full book, sets out dynamic and static options for epistemic modality (PDF pp. 1, 2, 4, 12, 16, 17; §§2–6, especially §6.2 on order) (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). The same material qualifies the inference in ways that bear on this paper. On its treatment, discourse-order effects do not by themselves establish semantic order effects within a sentence, and claims about the Boolean fragment are restricted (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). The diagnostic use of discourse order is therefore an established move, and the survey sets explicit modern limits on carrying it over to embedded conjunctions. The present paper should not be read as introducing order sensitivity as a new diagnostic.

The read literature also contains alternatives that challenge whether information sensitivity requires a separate semantic parameter or a nonclassical semantics. One such alternative is a pragmatic competition account of contrasts in modal embedding. It is read in its publisher proceedings version (PDF pp. 1, 2; abstract and introduction), and on its account modal embedding is not automatically evidence that uniquely favours one semantic family (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). A critical notice, read in an author manuscript (PDF pp. 1, 2), directly compares dynamic semantics with dynamic propositionalism. No neutral resolution of that comparison is inferred from it here (Willer 2023, PDF pp. 1, 2; §The Plot; critical notice of Stojnić). A formal account read as a 13-page extended abstract, and not either of its author's 2024 journal papers, formalizes stable acceptance and worldly information. It distinguishes acceptance relative to information states from pointwise truth (PDF pp. 1–3, 7–11; §§2–5) (Hawke 2023, PDF pp. 1, 2, 3, 7, 8, 9, 10, 11; §2–5, especially Stable Acceptance Semantics). These works show that relating epistemic modals to information states is already developed, and that whether the relation must be encoded in the semantics is contested.

One stronger nearby claim, that a static possible-worlds semantics treating sentence sequences as conjunctions cannot explain order-sensitive discourse, is also only partially addressed by these works. Rejecting all static semantics would require addressing pragmatic explanations and symmetry in embedded contexts (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). The present paper makes no such claim. There are also access gaps. A 2022 paper titled "Epistemic Contradictions Do Not Threaten Classical Logic" was identified only through metadata and could not be read, so it is recorded as an access gap rather than used as support. The same holds for other selected works unavailable to the search (Literature-currency search 2026, 2018–2026 search). Within these bounds, the contribution is

modest: an explicit reconstruction of one route from two dynamic-semantic commitments and one discourse contrast to an adequacy condition, together with its critical questions and limits. That thesis is partially anticipated by the read works cited above.

8. Conclusion

This paper has argued that an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The support comes from two commitments of dynamic semantics, namely that conjunction is interpreted sequentially and that might is a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18), and from the contrast in which the information that it is not raining, once added, is no longer consistent with the information that it might be raining, while the opposite order raises no problem (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Read as a sign, the contrast is explained by those commitments, and the explanation runs through the information state against which might is tested.

The conclusion is a demand on an account, and its wording should be kept exact. It requires that information states figure in modal evaluation. It does not require that worlds be removed. It holds conditionally on the stated premises, and it does not claim that every epistemic modal behaves as might does in the not-raining case. Whether other epistemic modals, other sequences and other modalities show the same pattern is left open. The condition defended here is what the stated commitments and the stated contrast support, and no more.

References

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